

SURNAME & INITIAL: I SITHOLE

STUDENT NUMBER: 47255242

CMPG 325: CLIENT DESIGN REVIEW

1. CLIENT REQUIREMENTS

Project Information

Project ID: CMPG325-2026-136

Client ID: CLI-136

Organization: Lenong Panelbeaters (Rustenburg)

Industry: Automotive

Network addressing block

The network has been allocated the IPv4 address block: **192.168.57.0/24**

This address block must be used for the network design and IP addressing plan.

Networking Challenge

The assigned networking challenge is DNS – Internal Name Resolution Service. The network needs an internal DNS solution that allows devices within the organization to resolve network hostnames.

Service availability requirement

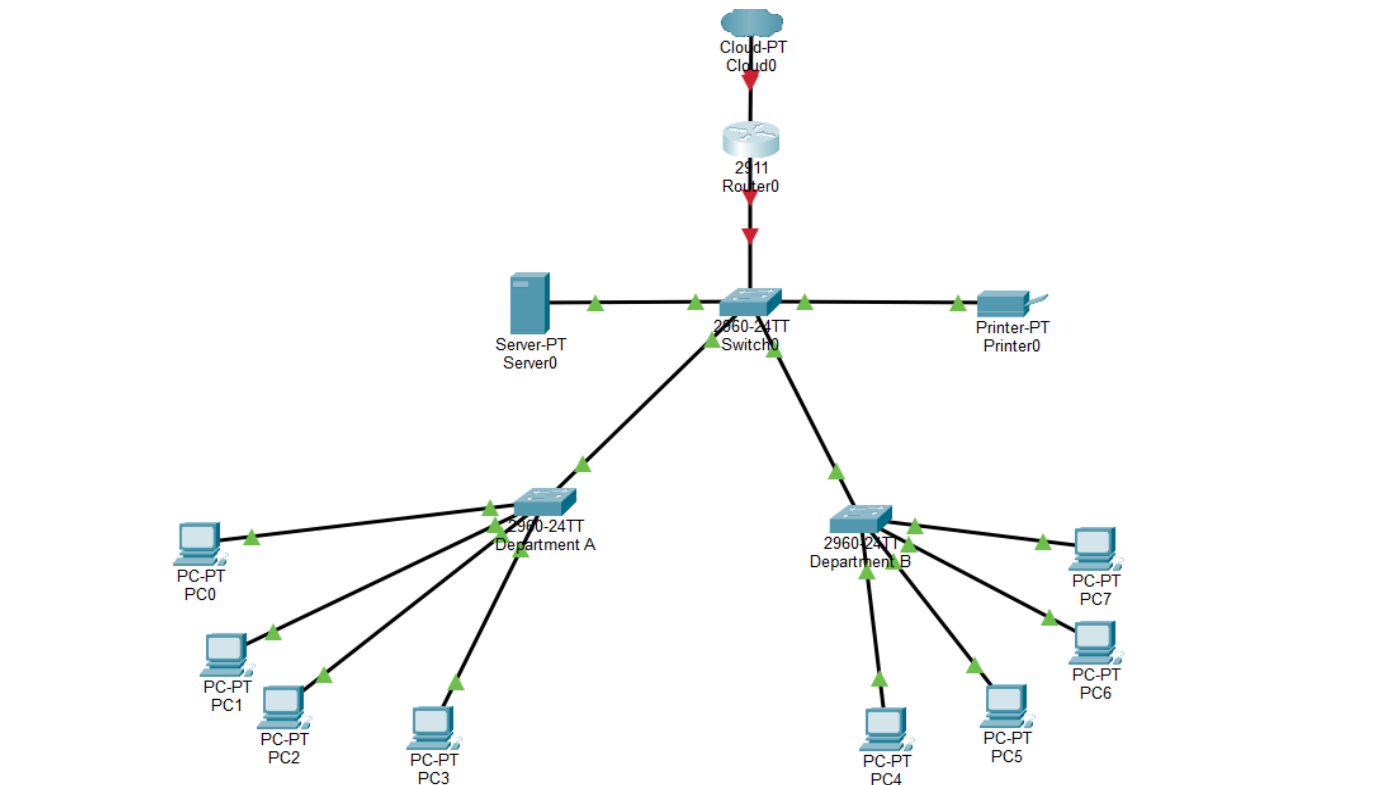
- File services
- Print services
- Application services

must remain available during business hours.

Change request-CR8

A shared printer zone must be provided to serve two departments that currently cannot print. The network design must therefore allow both departments to access the shared printing service.

2. PHYSICAL TOPOLOGY



3.LOGICAL TOPOLOGY

The logical topology for **Lenong Panelbeaters (Client ID: CLI-136)** is designed to ensure reliable connectivity, internal DNS resolution, and shared printing services for two

departments. The topology is structured around a star design, with a central distribution switch connecting all VLANs and critical services.

Core Layer

- Core Router: Provides connectivity between the internal LAN and the ISP.
- Configured with sub-interfaces for inter-VLAN routing (router-on-a-stick).
- Ensures that Department A, Department B, Servers, and the Printer Zone can communicate securely.

Distribution Layer

- Distribution Switch: Central switch connecting all access switches and server VLAN.
- Handles VLAN segmentation and forwards traffic to the router.
- Provides redundancy for critical services (file, print, application).

Access Layer

- Department A VLAN (**VLAN 10**): Connected via an access switch. Hosts PCs, printers, and phones for staff.
- Department B VLAN (**VLAN 20**): Connected via another access switch. Hosts workshop PCs and phones.
- Both VLANs are isolated but can communicate through the router for shared services.

Server VLAN (VLAN 30)

- Dedicated to hosting critical services:
 - DNS Server (internal name resolution).
 - File Server (document storage).
 - Application Server (business applications).
 - Print Server (manages shared printing).
- Ensures service availability during business hours.

Shared Printer Zone (VLAN 40 – CR8)

- Contains the shared printer connected to the Print Server.

- Accessible by both Department A and Department B through inter-VLAN routing.
- Implements the client's change request (CR8) by enabling shared printing functionality.

4. IP ADDRESSING PLAN

Assigned block: **192.168.57.0/24** (256 usable addresses).

Subnet	CIDR	Range	Purpose
192.168.57.0	/26	192.168.57.1 – 192.168.57.62	Department A VLAN (VLAN 10)
192.168.57.64	/26	192.168.57.65 – 192.168.57.126	Department B VLAN (VLAN 20)
192.168.57.128	/27	192.168.57.129 – 192.168.57.158	Server VLAN (VLAN 30 – DNS, File, Application, Print)
192.168.57.160	/27	192.168.57.161 – 192.168.57.190	Shared Printer Zone (VLAN 40 – CR8)
192.168.57.191 – 192.168.57.254		Reserved for future	

5. INITIAL GITHUB REPOSITORY

[Isau19/CMPG325-2026-136_Lenong_Panelbeaters: CMPG 325 Computer Networks project for Lenong Panelbeaters \(Rustenburg\).](#)